Feasible normals, inadmissibility candidates, and transfers

Notes for the restricted bivariate normal experiment

1. Setup and the positive theorem

Let

$$X \sim N_2(\mu, I_2), \qquad \Lambda = \{\mu \in \mathbf{R}^2 : \mu_2 \le \mu_1\},$$

with

$$\Lambda_0 = \{\mu \in \Lambda : \mu_1 \le 0\}, \qquad \Lambda_1 = \{\mu \in \Lambda : \mu_1 > 0\}.$$

A nonrandomized test is written as

$$\phi_A(x) = I\{x \notin A\},$$

where A is the acceptance region. For a closed convex set A, write

$$h_A(a) := \sup_{x \in A} a'x$$

for its support function. In the present model, the nonnegative feasible-normal wedge is

$$\mathcal{F} := \{a \in \mathbf{R}^2 : 0 \le a_2 \le a_1\} = \mathrm{cone}\{(1,0),(1,1)\}.$$

The condition $a \in \mathcal{F}$ says that a finite face of A points toward a mean direction that can be reached while respecting $\mu_2 \le \mu_1$ and without pointing against coordinatewise monotonicity.

Theorem 1 (Feasible-normal sufficiency)**.** *Let $A \subseteq \mathbf{R}^2$ be nonempty, closed, and convex. Suppose*

$$P_0(X \in A) = 1 - \alpha, \qquad 0 < \alpha < 1,$$

and suppose every nonzero direction in which A has finite support is feasible:

$$h_A(a) < \infty, \quad a \neq 0 \qquad \Longrightarrow \qquad a \in \mathcal{F}.$$

Then the test $\phi_A(x) = I\{x \notin A\}$ has level α for testing Λ_0 against Λ_1 and is α-admissible against all level-α tests, including randomized competitors.

Proof. First, the feasible-normal condition implies that A is a lower set. Indeed, let $x \in A$ and let $y \le x$ coordinatewise. If $y \notin A$, the separating hyperplane theorem gives a nonzero vector a such that

$$a'y > \sup_{z \in A} a'z = h_A(a).$$

Hence $h_A(a) < \infty$, so $a \in \mathcal{F}$ and in particular $a \ge 0$ coordinatewise. But then $a'y \le a'x \le h_A(a)$, a contradiction. Therefore $y \in A$.

Since A is a lower set, the rejection region A^c is coordinatewise upper. If $\mu \in \Lambda_0$, then $\mu \le 0$ coordinatewise. Coupling $X = \mu + Z$ and $Z \sim N_2(0, I_2)$ gives $\mu + Z \le Z$ coordinatewise, and therefore

$$P_\mu(X \notin A) \le P_0(Z \notin A) = \alpha.$$

Thus the test has level α, with exact size attained at the finite null point $\mu = 0$.

It remains to verify the Lehmann–Romano ray condition. In the present normal experiment the natural parameter is $\theta = \mu$. Suppose

$$A \cap \{x : a'x > c\} = \varnothing.$$

If $a = 0$, the condition is vacuous because one may choose any fixed point in Λ_1. If $a \neq 0$, then $h_A(a) \leq c < \infty$, so by assumption $a \in \mathcal{F}$. Hence $a_1 > 0$ and $0 \leq a_2 \leq a_1$. Therefore, for every $\lambda > 0$,

$$\lambda a \in \Lambda_1.$$

The required ray is obtained by taking $\theta^* = 0$ and any sequence $\lambda_n \to \infty$.

The Lehmann–Romano convex-acceptance-region result then gives d-admissibility, and because the level is α and is attained at the finite null point 0, its corollary gives α-admissibility. The conclusion is against all level-α competitors, including randomized competitors. □

Remark 1 (Examples covered by the theorem). The short test $I\{x_1 > c_s\}$ is covered, since $A_s = \{x_1 \leq c_s\}$ has finite normal $(1, 0)$. The diagonal-sum test $I\{x_1 + x_2 > b\}$ is covered, since its finite normal is $(1, 1)$. The gated-sum family

$$A_{c,b} = \{x_1 \leq c,\ x_1 + x_2 \leq b\}, \qquad \phi_{c,b}(x) = I\{x_1 > c \text{ or } x_1 + x_2 > b\},$$

is covered, since its finite normals lie in cone$\{(1, 0), (1, 1)\}$.

2. Tests failing the feasible-normal condition

The theorem is a sufficient condition, not an iff statement. A finite normal outside $\mathcal{F}$ should be treated as an inadmissibility diagnostic: it identifies a face that spends rejection probability in a direction that cannot be probed by restricted alternatives. The following tests are therefore candidates for inadmissibility; some are already known to be inadmissible.

Test or family	Acceptance region in X-coordinates	Status and reason
X_2-only test	$\{x_2 \le q\}$	Fails because the finite normal is $(0,1) \notin \mathcal{F}$. Known inadmissible: it is dominated by the corresponding X_1-only test.
Difference test	$\{x_2 - x_1 \le q\}$	Fails because the finite normal $(-1,1)$ points toward the deleted direction $\mu_2 - \mu_1 > 0$. Known inadmissible in the restricted problem.
Long/max test	$\{x_1 \le c_\ell,\ x_2 \le c_\ell\}$	Fails because of the vertical face with normal $(0,1)$. Known inadmissible globally; dominated by a gated-sum test.
Finite ordered rectangles	$\{x_1 \le c_1,\ x_2 \le c_2\}$, $c_1 \le c_2 < \infty$	Same failure as the long test: the normal $(0,1)$ is infeasible. Known admissible only relative to rectangular competitors, not globally.
AS max / ordered-largest statistic with one selected violation	$\max\{x_1^+, x_2^+\} \le r$, equivalently $x_1 \le r,\ x_2 \le r$	This is the long rectangle in the two-inequality identity-covariance case. Known globally inadmissible.
AS MMM statistic S_1	$(x_1^+)^2 + (x_2^+)^2 \le r^2$	Fails: the rounded boundary has normals proportional to (x_1^+, x_2^+), including normals with $x_2 > x_1$, and the face $x_2 = r$, $x_1 \le 0$, has normal $(0,1)$. Candidate for inadmissibility.
AS QLR statistic S_2, $\Omega = I_2$	Same as MMM: $(x_1^+)^2 + (x_2^+)^2 \le r^2$	In the simple identity-covariance, two-inequality case, QLR coincides with MMM. Candidate for inadmissibility. For $\Omega \ne I_2$, the geometry is a projected ellipsoidal distance and should be studied separately.

Here $x_j^+ = \max\{x_j, 0\}$ and the sign convention is the moment-inequality convention $x_j = -m_j/\sigma_j$, so that larger x_j means stronger violation of inequality j. For Andrews–Soares, the

MMM statistic is

$$S_1(m, \Sigma) = \sum_{j=1}^{p} [m_j/\sigma_j]_-^2$$

when there are no equalities, and the QLR statistic is the distance from m to the nonnegative orthant in the Σ^{-1} norm. Thus, for $p = 2$, no equalities, and $\Sigma = I_2$, both give

$$S(-x, I_2) = (x_1^+)^2 + (x_2^+)^2.$$

The max statistic with one selected violation gives

$$S(-x, I_2) = \max\{(x_1^+)^2, (x_2^+)^2\},$$

and hence the long rectangular acceptance region. Andrews–Soares Assumption 1 gives coordinatewise monotonicity after this sign change, but it does not impose the feasible-normal condition.

3. A formal transfer criterion

The feasible-normal theorem gives a sufficient condition for admissibility. A complementary way to prove inadmissibility is to construct a null-mass-preserving transfer of rejection probability toward points that have uniformly larger likelihood ratios under the restricted alternatives.

Every $\mu \in \Lambda_1$ can be written as

$$\mu = (u, u - v), \qquad u > 0, \qquad v \geq 0.$$

Relative to P_0, the likelihood ratio is

$$L_\mu(x) = \exp\left\{u(x_1 + x_2) - vx_2 - \frac{1}{2}(\mu_1^2 + \mu_2^2)\right\}.$$

Thus likelihood ratios are increasing in $x_1 + x_2$ and decreasing in x_2. Define the restricted-likelihood order by

$$x \succeq_\Lambda y \quad \Longleftrightarrow \quad x_1 + x_2 \geq y_1 + y_2 \quad \text{and} \quad x_2 \leq y_2.$$

Proposition 1 (Transfer domination criterion)**.** *Let ϕ and ψ be nonrandomized tests with rejection regions R_ϕ and R_ψ. Define the gain and loss sets*

$$B := R_\psi \setminus R_\phi, \qquad D := R_\phi \setminus R_\psi.$$

Suppose

$$P_0(B) = P_0(D) < \infty$$

and suppose there exists a coupling Π of the two finite measures $P_0(\cdot \cap B)$ and $P_0(\cdot \cap D)$ such that

$$\Pi\{(x, y) : x \succeq_\Lambda y\} = P_0(B).$$

Then

$$E_\mu[\psi(X)] \geq E_\mu[\phi(X)] \qquad \textit{for every } \mu \in \Lambda_1.$$

If, for some $\mu \in \Lambda_1$,

$$\Pi\{(x, y) : \mu' x > \mu' y\} > 0,$$

then the inequality is strict at that μ.

Proof. Since the two tests differ only on $B \cup D$,

$$E_\mu[\psi(X)] - E_\mu[\phi(X)] = P_\mu(B) - P_\mu(D).$$

Using $L_\mu = dP_\mu/dP_0$ and the coupling Π,

$$P_\mu(B) - P_\mu(D) = \int \{L_\mu(x) - L_\mu(y)\}\, d\Pi(x, y).$$

If $x \succeq_\Lambda y$, then

$$\mu'(x - y) = u\{(x_1 + x_2) - (y_1 + y_2)\} + v(y_2 - x_2) \geq 0,$$

so $L_\mu(x) \geq L_\mu(y)$. This proves weak domination. The strict statement follows from the same display when $\mu' x > \mu' y$ on a set of positive Π-measure. □

Remark 2 (How the rectangle-to-gated improvement fits). For a finite ordered rectangle $A_c = \{x_1 \leq c_1, x_2 \leq c_2\}$, the gated replacement keeps the gate $x_1 \leq c_1$ and replaces the vertical face by

$$A_b = \{x_1 \leq c_1,\ x_1 + x_2 \leq b\}, \qquad P_0(A_b) = P_0(A_c).$$

The new rejection set gains points with larger $x_1 + x_2$ and lower x_2, and gives up points with smaller $x_1 + x_2$ and higher x_2. Thus the gain-loss transfer is ordered by $\succeq_\Lambda$, which explains the domination. This is the template to try next for the MMM and QLR candidates.

4. Action plan

1. Treat Theorem 1 as the positive result: it already gives globally admissible tests that genuinely use X_2, including the gated-sum and pure diagonal-sum tests.
2. Treat failure of $\mathcal{F}$ as a diagnostic, not yet as a theorem. The conjectural converse is: among closed convex exact-size lower tests, global admissibility is equivalent to having all finite support directions in $\mathcal{F}$.
3. For each failing candidate, try to construct B and D satisfying Proposition 1. This is already done for finite rectangles. The next natural target is the AS MMM/identity-QLR rounded set.
4. For MMM/QLR, look first at the boundary arc where $x_2 > x_1 > 0$ and at the vertical face $x_2 = r$, $x_1 \leq 0$. These are the pieces with infeasible normals. A plausible dominator should shave mass from those pieces and move it toward larger $x_1 + x_2$ and lower x_2 while preserving null probability.

5. Postpone $\Sigma \neq I_2$ until the identity case is settled. For nonidentity covariance, the LR ray direction is a natural-parameter direction and the mean direction is Σa, so the feasible-normal condition must be transformed through Σ.

References

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[2] Lehmann, E. L. and Romano, J. P. (2005). *Testing Statistical Hypotheses*. Third edition. Springer.